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A palaeomagnetic study of part of the Lewisian complex, north-west Scotland

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SUMMARY

Directions of magnetization have been measured in dykes and gneisses of the Central Zone of the mainland Lewisian and in the Laxford granite sheets. Nine Scourie dykes from an area around Lochinver have yielded closely grouped directions of magnetization giving a mean palaeomagnetic pole at 37·3°N, 274·8°E (dp = 6·9°, dm = 9·6°). The direction for a Scourian granulite falls very close to the mean for the dykes.

A regional difference in magnetic stability has been discovered and this, together with evidence concerning variations in the opaque minerals, has formed the basis of an interpretation in terms of metamorphic history. Stable magnetization was probably imposed in the late Laxfordian metamorphism at about 1800 Ma. Upon closure of the Atlantic Ocean there is good agreement with North American apparent polar wander paths.

THE LEWISIAN of north-west Scotland is a metamorphic complex composed mainly of banded gneisses but incorporating dykes, pegmatites and granite sheets as minor intrusive constituents. Three zones (Fig. 1) have been recognized on the mainland (Peach *et al.* 1907, Sutton & Watson 1951). The Central Zone consists primarily of gneisses in the granulite facies formed during the Scourian metamorphism; in the Northern and Southern Zones the gneisses were reworked in amphibolite facies in the Laxfordian metamorphism. These zones were originally established with the aid of the dykes which are often undeformed and only slightly metamorphosed in the Central Zone but profoundly altered elsewhere. Both the Scourian and Laxfordian are now regarded as polyphase episodes responsible for repeated heating and deformation. Lewisian events relevant to this study are summarized in Table 1 (based upon Evans & Lambert 1974).

1. Palaeomagnetic sampling & results

This work began on little-altered Scourie dykes in the belief that these would be most likely to have retained pre-Laxfordian magnetization. Later work included studies of the country rocks and granites. A two-stroke drill was used to collect cores 2·5 cm in diameter and *c.* 6 cm long. They were orientated using a Brunton compass in conjunction with topographic sighting. A total of 775 samples was collected from 28 dykes, 7 gneisses and 3 granites, the sampling localities lying within and at the northern border of the Central Zone. A specimen 2·5 cm long was subsequently sliced from each sample for the measurement of magnetization, the remainder being available for the preparation of thin and polished sections.

An astatic magnetometer with a maximum sensitivity of 2×10^{-6} emu/cm^3 per cm deflexion was used to measure the natural remanent magnetization (NRM) of each specimen and the directions of magnetization after partial alternating field demagnetization. This treatment was applied to remove unstable secondary

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components and to test the stability of any underlying magnetization. In unstable rocks the magnetic vector turns through a large angle at each demagnetizing stage, the intensity falls sharply and there is little or no tendency for the directions of the vectors of individual specimens to form a cluster. The converse is true for stable rocks; in these the grouping improves with each demagnetizing step until optimum grouping is achieved. A wide range of stability was found as illustrated in Figs. 2, 3, 4. A typical example of unstable material is provided by Dyke 1, the well-known 'graveyard dyke' at Scourie, which was heavily sampled; the specimens were so unstable that the direction of the NRM could be reversed simply by banging them with a hammer. Moreover alternating field demagnetization has a randomizing effect, producing approximately equal numbers of normally and reversely magnetized specimens (Fig. 4(*b*)).

Stable remanence was recovered from ten dykes and one granulite whereas the remaining units were unstably magnetized. The results for these eleven units are set out in Table 2 and are based upon the demagnetizing fields at which the grouping of directions is best. The positions of the sampling sites are shown in Fig. 5. All the dykes sampled are within the Central Zone which was geologically fairly stable

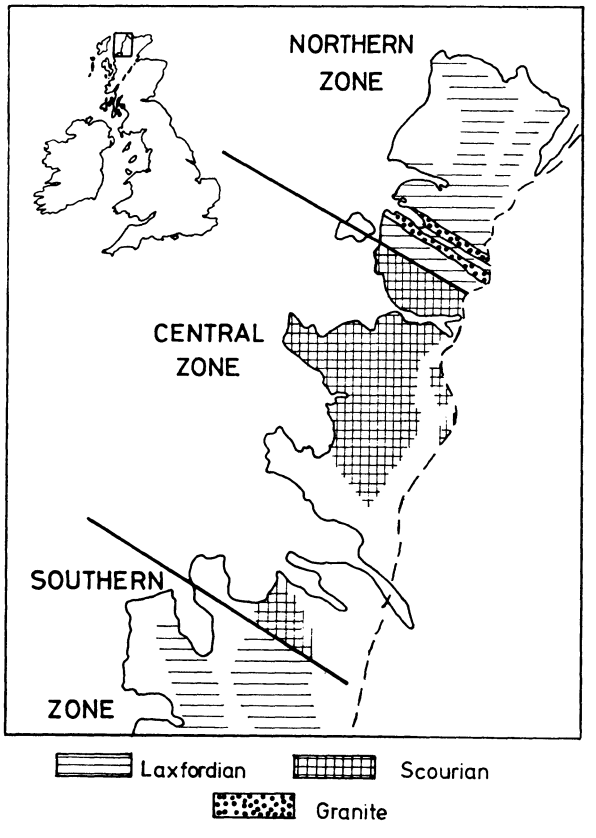


FIG. 1. The metamorphic zones of the mainland Lewisian (after Ramsay 1969).

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in Laxfordian times. However, magnetic stability displays a distinct regional variation. All but one of the stably magnetized rock units lie in the 'stable' region near Lochinver, the sole exception being Dyke 7 (q.v.). It may be seen (Fig. (6*a*)) that the mean directions of magnetization of nine of the dykes form a tight group; moreover the direction for the granulite gneiss G6 (unfortunately the only granulite sampled in this region) falls within this cluster. On the other hand the direction for Dyke 7 is distinctly different. In the 'unstable' region all gneisses, granites and all the other dykes proved unstable. Their behaviour is very similar to Dyke 1 and so stable specimens were rare and (apart from Dyke 7) no palaeomagnetic directions emerged from this region.

An interesting example of intermediate stability is represented by Dyke 11 from which only 11 specimens out of 57 proved magnetically stable (using the dyke stability criterion of Beckmann 1973). This dyke is both the most northerly (except Dyke 7) and the most easterly site from which a palaeomagnetic direction was recovered. The Moine Thrust zone (lying only 1 km away) is visible from the dyke and the lowest major thrust displays Lewisian gneisses on top of Cambrian rocks. Some disturbance of U-Pb ages of zircons in granulites in the vicinity of Dyke 11 has been noted and attributed to the Caledonian orogeny (Pidgeon & Bowes 1972). It is therefore possible that Caledonian events contributed to the instability of Dyke 11, although in general their effects upon the portion of the Lewisian covered by the sampling are considered negligible.

It is not claimed that all rocks in the so-called 'stable' region are in fact magnetically stable (three dykes were discovered here which were highly unstable) but that this is a region where stable units may be found. On the other hand it does seem unlikely that any magnetically stable units can be found amongst the typical dykes and gneisses in the 'unstable' region. The thin sections were kindly examined by Professor J. V. Watson who devised four categories of slight alteration of the dykes. However, there appeared to be no local correlation between the extent of alteration and the palaeomagnetic results.

TABLE 1. *Lewisian Chronology*

Event	Time before present, Ma.
Slow cooling of the crust, perhaps with distinct post Late Laxfordian episodes, particularly at 1,400 Ma in certain localities	1,750 - 1,400
End of the Late Laxfordian	c. 1,750
Peak of Late Laxfordian amphibolite facies metamorphism	⩾ 1,850
Early Laxfordian incipient granulite facies metamorphism	undated
Intrusion of the Scourie dykes	2,200 - 1,900
Inverian (Late Scourian) amphibolite facies metamorphism	2,310 - 2,200
Close of the Early Scourian (Badcallian) metamorphism	⩾ 2,600
Peak of the Early Scourian granulite facies metamorphism	c. 2,900

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2. Interpretation

The directions of magnetization are unlike any known Phanerozoic directions from Britain and are, therefore, probably Precambrian in age. The grouping of directions and common polarity of the nine stable dykes imply that they were magnetized over a limited time. Moreover the result for the granulite G6 suggests that the country rocks were magnetized simultaneously. The demagnetizing behaviour of this granulite is very similar to that of the stable dykes (Fig. 3), indicating that the magnetic material has comparable coercivities. This could be explained by an experience common to the dykes and country rocks such as a metamorphic episode. It will be suggested that the late Laxfordian metamorphism is the most likely event. This view is based on some geochronological evidence and on arguments involving temperature, shock and opaque mineralogy.

(A) GEOCHRONOLOGY

Several Scourie dykes from the Assynt area were dated by Evans & Tarney (1964)

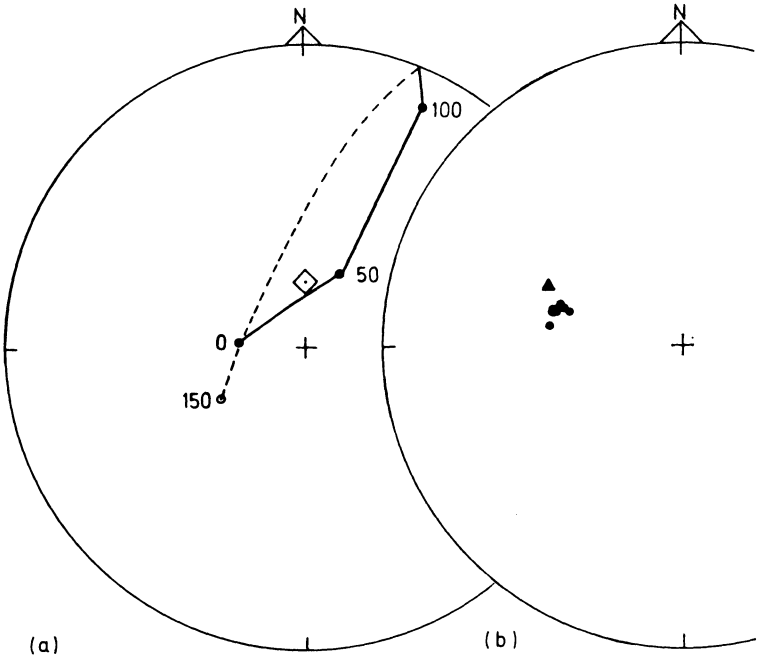


FIG. 2. Directions of magnetization after magnetic cleaning for (a) a specimen from the unstable Dyke 12 (the numerals represent the peak demagnetization fields in oersteds); (b) a specimen from the stable Dyke 26 (directions for demagnetizing fields of 0, 25, 50, 75, 100, 125, 150, 200, 250, 300, 400, 500 oersteds are shown; the average angular change between each step is 2.7°). Diamond symbol indicates direction of present axial dipole field; triangle shows mean direction for all samples from the dyke.

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using the K/Ar method and in one case the Rb/Sr technique. The best estimate for the date of intrusion was found to be 2200 Ma with a spread of apparent ages down to about 1950 Ma. In the present study, three of the group of nine dykes having similar directions of magnetization had already been dated by Evans & Tarney. Dyke 26 gave values of 3860 Ma and 3800 Ma, Dyke 27 gave 1950 Ma whilst Dyke 28 yielded the dates 2740 Ma and 1800 Ma.

The ages for Dyke 26 and the older one for Dyke 28 are excess argon dates and serve to emphasize the need for caution when correlating isotopic ages with palaeomagnetic directions in Precambrian terrains. The 1800 Ma date was obtained from a point on the dyke very close to a Laxfordian hot shear and it was from this location that the palaeomagnetic samples were taken. This age reflects local argon loss at 1800 Ma (or later).

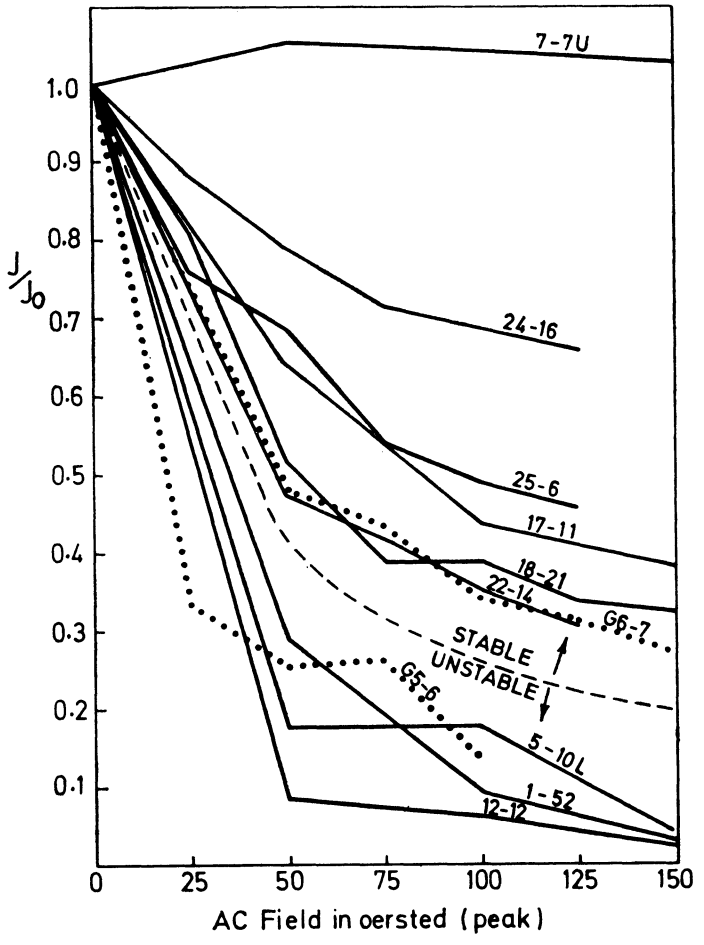


FIG. 3. Alternating field demagnetization curves. Specimens have been chosen to be as representative as possible of the rock units from which they were collected. Full lines refer to dyke specimens, dotted lines to gneisses.

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Dyke 27 is fine-grained, cuts early types of dyke, is rather unaltered and the gneiss margins show brittle fracturing. On these grounds it is regarded as a late but pre-Laxfordian member of the dyke suite, intruded into cool rock (Tarney 1963, Evans & Tarney 1964, Tarney 1973). The date of 1950 Ma is considered close to the true time of emplacement.

The date for Dyke 27 and the lower date for Dyke 28 are near those assigned to the climax of the late Laxfordian (Lambert & Holland 1972), making it reasonable to suppose that these dykes acquired their magnetism then. In spite of the indications that Dyke 27 was intruded after the typical dykes, exemplified by the other eight members of the group of stable dykes, its direction of magnetization falls close to their mean. This suggests the possibility that the whole group was last magnetized after the time of intrusion, perhaps in the late Laxfordian.

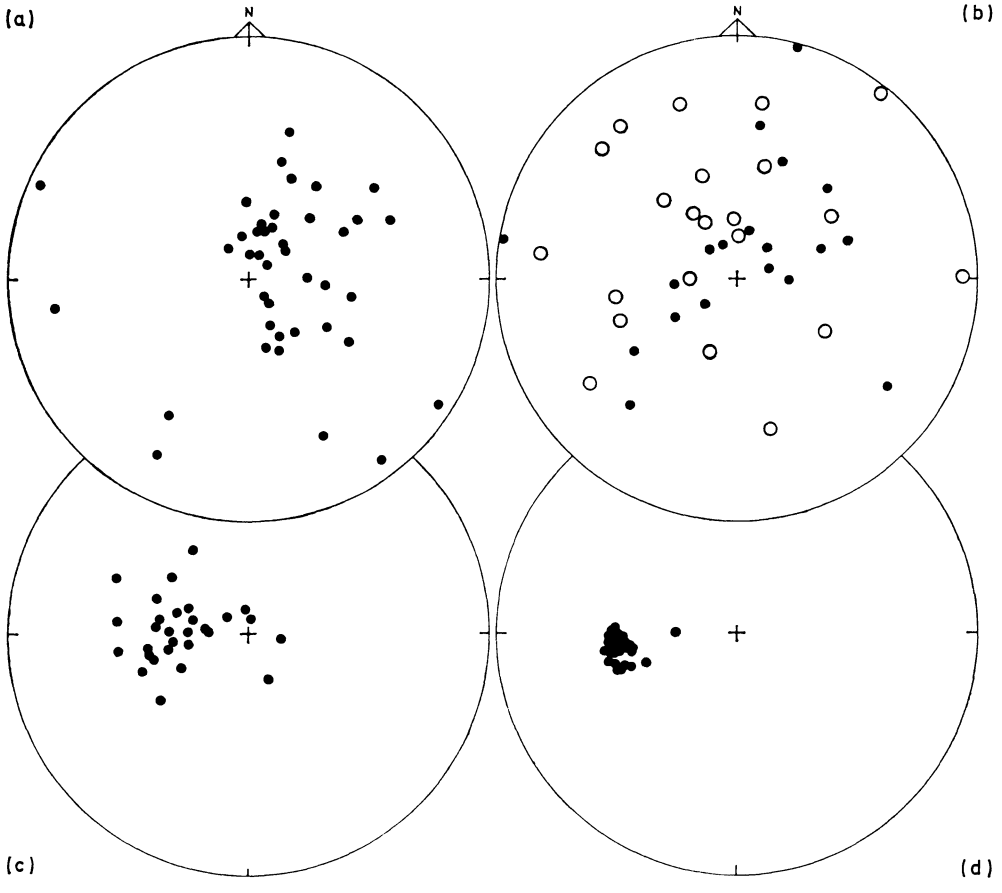


FIG. 4. Directions of magnetization for all samples from the unstable Dyke 1 and the stable Dyke 18. (a) N.R.M. of Dyke 1, (b) Dyke 1 after magnetic cleaning in a peak field of 150 oersteds, (c) N.R.M. of Dyke 18, (d) Dyke 18 after magnetic cleaning at 150 oersteds. Open circles denote directions in the upper hemisphere, solid circles directions in the lower hemisphere.

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(B) TEMPERATURE

The late Laxfordian metamorphism was characterized by regional variations in intensity, being more intense in the Northern and Southern Zones where amphibolite facies minerals are fully developed; here the temperature might have been about 650–700°C. In contrast, the Central Zone was largely unmodified, having experienced only patchy mineralogical and structural effects. For this reason, and on account of the survival of early Scourian dates, the temperature in the middle of the Central Zone might have been only 400–500°C in the late Laxfordian. Thus during this metamorphic episode the northern half of the sampling area, in which unstably magnetized rocks are typical (Fig. 5), would have been hotter than the stable region. Magnetic stability may therefore be linked with variation in the intensity of the late Laxfordian mobile episode. This apparent association in time of the magnetization with the late Laxfordian implies either that the stable magnetization was acquired in this metamorphism, but could not establish itself to the north, or that in the 'stable' region previous magnetization persisted whereas it was obliterated in the 'unstable' region by the higher temperatures. Observations of the opaque minerals have furnished a theory which helps to resolve these alternatives and have suggested mechanisms for the magnetization.

(C) OPAQUE MINERALOGY

Polished rock sections were examined using a Reichert ore microscope to establish the modes of occurrence of magnetite and ilmenite and the presence or absence of sulphides. Ilmenite in Scourie dykes and gneisses from the unstable region occurs as well-formed grains taking an excellent polish and often lying next to magnetite grains.

TABLE 2. *Stably magnetized rock units*

Rock Unit	H	N	N'	Mean direction of magnetization				Virtual geomagnetic pole			
				Dec.	Inc.	k	α_{95}	Lat°N	Long°E	dp	dm
Dyke 7	100	36	32	106.1	1.0	35	4.4	-7.9	70.8	2.2	4.4
Dyke 11	50	57	11	264.7	55.6	70	5.5	27.5	290.0	5.6	7.9
Dyke 17	50	24	14	301.1	62.1	45	6.0	51.4	268.8	7.2	9.3
Dyke 18	150	30	29	262.0	50.7	166	2.1	22.3	288.8	1.9	2.8
Dyke 22	125	25	16	297.1	55.4	18	8.9	43.9	264.6	9.0	12.7
Dyke 24	100	17	11	294.0	49.6	200	3.2	38.0	262.6	2.9	4.3
Dyke 25	25	24	6	298.7	57.4	66	8.3	46.3	265.1	8.9	12.2
Dyke 26	75	33	27	293.1	50.3	184	2.1	38.0	263.9	1.8	2.8
Dyke 27	50	20	6	284.0	63.8	36	11.3	44.0	283.7	14.2	17.9
Dyke 28	125	29	29	272.4	46.6	202	1.9	24.6	278.5	1.6	2.4
Granulite,G6	125	22		290.8	59.5	25	6.3	43.8	273.8	7.2	9.5

H is the optimum demagnetizing field in (peak) oersted. N is the number of specimens measured. N' is the number of stable specimens. The Declination of the mean direction of magnetization is measured in degrees East of true North and Inclination is measured from the horizontal and regarded as positive downwards. The statistics are those of Fisher (1953); k is a measure of the precision of grouping of directions and α_{95} is the semiangle of the cone of 95% confidence about the mean direction. The virtual geomagnetic pole is calculated from the mean direction; dp and dm are the semiaxes of the oval of confidence about this pole at the 95% probability level, dp in the colatitude direction and dm perpendicular to it.

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In contrast two interesting features have been discovered in ilmenite from the stable region. First, irregular blades (Pl. 1A) in ilmenite have been seen in many of the stable dykes and in the stable granulite. It has been established by electron microprobe analysis that no compositional difference exists between blades and host grains. Secondly, a different kind of blade (Pl. 1C) has been seen in ilmenite in two of the stable dykes (26, 28). These may be described as 'white blades.' They are straight and parallel, often terminating within a grain; the parallelism may

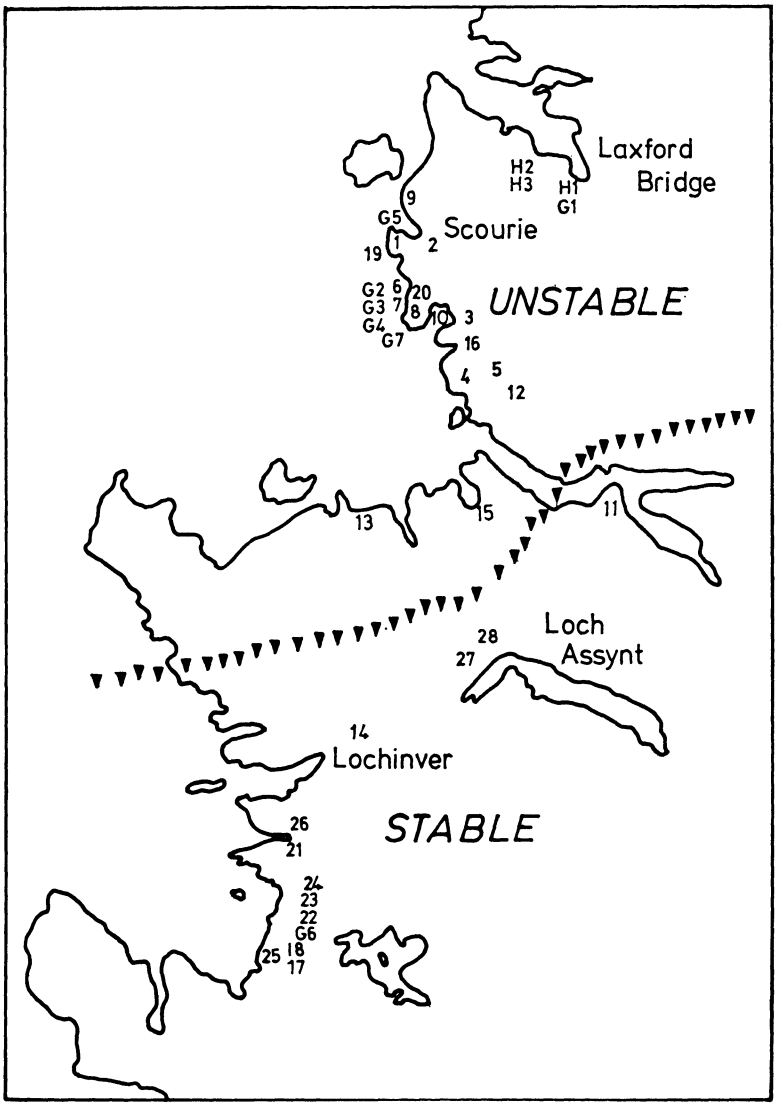


FIG. 5. Location of sampling sites; the prefix G indicates a gneiss, H denotes granites, whereas numbers without prefixes represent dykes. This map also illustrates the division into stable and unstable regions; the latitude of their boundary is thought to be approximately correct but its course is as yet uncertain.

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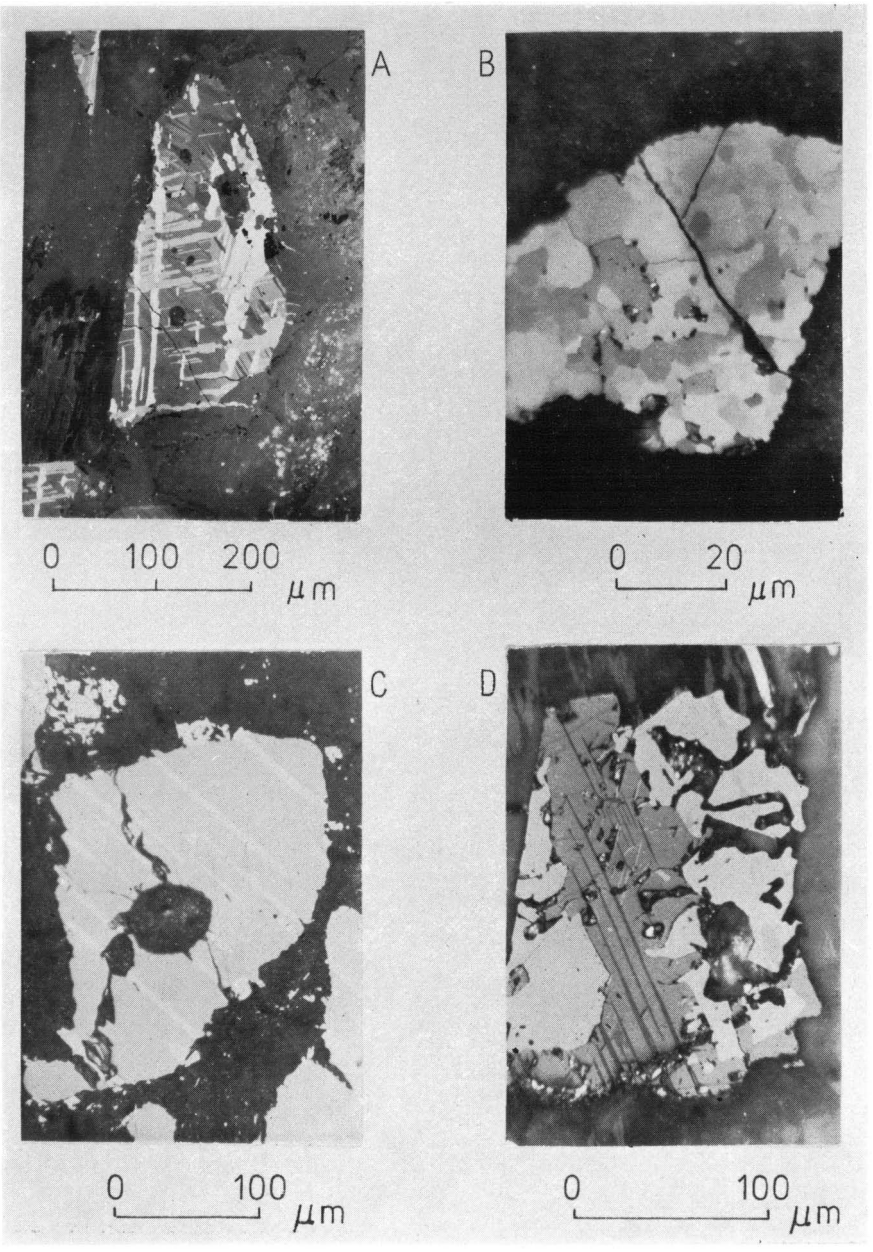


PLATE I

A. Irregular blades in ilmenite.
B. Mosaic texture in ilmenite.
C. White blades in ilmenite.
D. Ilmenite lamellae in a magnetite grain (on each side of which lie separate grains of ilmenite).

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frequently be traced over several grains. In contrast with the 'irregular blades' there does exist a compositional difference between blades and host grain. The blades are essentially haematite but contain substantial amounts of titanium and vanadium with a very little manganese. The host grain on the other hand contains much less vanadium but more manganese. The two dykes in which this feature occurs have high precision parameters ($k = 640$ before demagnetization for the first 22 cores collected in Dyke 28) and the direction of magnetization changes but little upon demagnetization. (Microprobe analyses kindly carried out under the direction of Dr R. Dearnley at the Institute of Geological Sciences.)

Shock effects in terrestrial ilmenite have been studied and compared with features seen in lunar ilmenite (Sclar *et al.* 1973). Both shock-induced mechanical twinning was found to occur and shock-induced unmixing of an ilmeno-haematite solid solution, which gave an orientated intergrowth of haematite in ilmenite. In lunar fragments displaying igneous textures none of the transparent minerals show evidence of shock damage whereas the ilmenite invariably displays twinning. Sclar *et al.* emphasize the potential value of ilmenite as a sensitive indicator of shock. It therefore seems likely that the features noted in the Lewisian ilmenite (at least the irregular blades) were also caused by shock. Their distribution is consistent with a late Laxfordian interpretation. In Pl. 1A the aggregates of small crystals (on top right hand side of the crystal) result from thermal annealing after the shocks. In a few examples such a mosaic texture fills the whole crystal indicating a still later stage in recrystallization (Pl. 1B). The absence of the blades in the unstable region is thought to imply complete recrystallization resulting from the higher temperatures prevailing there after Laxfordian shocks had ceased.

It is believed that in all the rocks examined the carrier of the magnetic remanence is magnetite. Although ilmenite does not carry remanence itself it is very

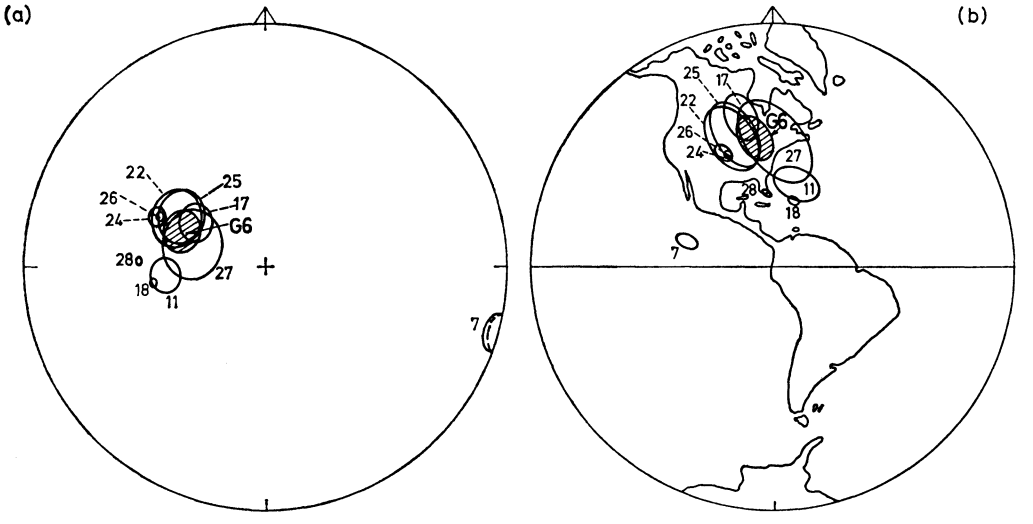


FIG. 6. (a) Mean magnetization directions for the 11 stable rock units.
(b) Virtual geomagnetic poles for the same rock units. Ovals of 95% confidence are shown in each case. The pole for Dyke 7 has been reversed.

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closely associated with magnetite. The observation of features in ilmenite which are probably linked with variations in Laxfordian temperatures hints at the possibility that the state of the magnetite may also be connected with this metamorphism.

Ilmenite lamellae in magnetite (Pl. 1D) are common in the late Dyke 27 but have not been seen in any of the other dykes. A possible explanation of this difference is that in the typical dolerites any ilmenite lamellae initially present were swept out of the magnetite grains by the prolonged high temperatures which they experienced.

(D) MECHANISMS OF MAGNETIZATION

Stable magnetization is widely associated with single domain grains whereas unstable magnetization may be expected from multidomain structures. However mechanisms have been put forward whereby multidomain grains behave like single domains. It has been argued that such behaviour arises from dislocations which cause local stresses in magnetite crystals (Verhoogen 1959, Stacey & Wise 1967, Kobayashi & Fuller 1968). The stability and intensity of remanence in magnetite have been found to depend critically upon the state of internal stress which is thought to impede wall movement by creating barriers to their motion (Lowrie & Fuller 1969, Schmidt 1973). As illustrations of these ideas, experiments on magnetite have shown (van Oosterhout & Klomp 1962) that grinding increases its coercive force whereas subsequent annealing restores lower values. Moreover in the present study a highly unstable specimen from a Laxford granite sheet could be rendered relatively stable by banging it with a hammer; also it has been shown (H. J. Mauritsch, personal communication) that the stability of basalts can be altered by banging.

These lines of evidence, with that from the ilmenite grains, make it reasonable to postulate that the stable magnetization in the Lewisian might be accounted for in terms of shocks, whose effects persisted in the stable region but were annealed in the unstable region. The stable magnetization may be a medium-temperature stress-induced remanence; however, the high precision parameters for four of the dykes suggest the sudden locking of the magnetization by specific shocks. This idea leads to an alternative mechanism: the shocks might have produced local high temperatures followed by the acquisition of thermo-remanent magnetization.

Whilst the magnetization could have occurred by these mechanisms there is so far no evidence that it did occur like this. Experiments are at present in progress using electron microscopy to test the hypothesis that there are more dislocations in magnetite and ilmenite in rocks from the stable than from the unstable region.

Suppose however that it is correct to interpret the magnetization in terms of shock; the most recent events involving shocks directly affecting dykes and country rocks would be of late Laxfordian age. This interpretation is favoured here because this is the only important phase affecting Dyke 27, although it is not yet possible to assign a precise age to the magnetization within the late Laxfordian episode. Nevertheless following Lambert & Holland (1972), it is tentatively proposed that the 'stable' region was last magnetized 1800 ($\pm$50) Ma ago. Thus it is suggested that the stable Scourie dykes were finally magnetized long after intrusion. Even

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in Palaeozoic times cases have been distinguished of rocks not acquiring their magnetism until a long time after emplacement, for example the Aberdeenshire gabbros (Sallomy & Piper 1973) apparently became magnetized about 20 Ma after intrusion into Dalradian terrain undergoing regional metamorphism.

(E) POLARITY

The common polarity of the nine dykes and the granulite (which lie as much as 24 km apart) is a striking feature. It seems likely that there were limited differences in the time that each of these ten units was magnetized but that they were all magnetized within the same polarity epoch. The Wind river dykes (Spall 1971a) of age $\geqslant$ 1880 Ma, the Marathon dykes, 1810 Ma (Fahrig *et al.* 1965) and dykes and gneisses from Central West Greenland (Beckmann & Mitchell, also Morgan, in preparation) all have the same polarity as the Scourie dykes. There are therefore some indications that there may have been a substantial episode of constant polarity at about 1800 Ma.

As far as the Lewisian is concerned, one mechanism which seems unlikely is acquisition of magnetization by very slow cooling during a period when the Earth's field was reversing itself at the same rate as at present. Fall of temperature between reversals might have amounted to less than 1°C, with the result that the rocks would fail to acquire any remanence. It is conceivable that the rocks near Scourie owe their instability to being above their Curie point at the time of the suggested constant polarity epoch during which the cooler rocks to the south acquired their magnetization.

3. Dyke 7

This dyke at Badcall near Scourie has a different direction of magnetization from the other stable units and is the only stable unit found in the 'unstable' region. It is not a normal Scourie dyke in that its chemistry (Tarney, unpublished) and mineralogy (O'Hara 1962) are unusual. It has been termed a 'norite' because it contains more hypersthene than clinopyroxene. The opaque mineralogy of this dyke is also different from the other Scourie dykes; the magnetite grains are generally less than 30 microns across, compared with some hundreds of microns in the typical dolerites. The opaque mineral assemblage is relatively simple, resembling that of Tertiary dykes in Mull (Ade-Hall, personal communication). This dyke has been dated at 1860 Ma where least altered and 1440 Ma where amphibolitized (Evans & Tarney 1964). If it is assumed that the magnetization was imposed at one of these times it seems a little more likely that this dyke was magnetized at 1440 Ma than at 1860 Ma on account of the nearly opposite polarity and because this region would have cooled later than the Assynt region.

4. Comparison with North American polar wander paths

Pole positions for the stable rock units are shown in Fig. 6(b). It is generally believed that the Atlantic Ocean was closed for at least part of the Phanerozoic.

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Europe can be restored to its predrift position by a clockwise rotation of 38·1° about a pole at 88·4°N, 27·7°E (Bullard *et al.* 1965). In the following discussion we assume that this has been done.

At least three Precambrian apparent polar wander (a.p.w.) paths have been published (Figs. 7, 8): Spall (1971b), Irving & Park (1972) (modified by McGlynn *et al.* 1974), and Piper (1974). There is disagreement between the first two curves prior to 1400 Ma. Although the curve of Irving & Park is based on some of the same poles as Spall's, as well as additional data, they have had to set aside certain results in order to present a simple curve. In particular the pole from the Wind River dykes (> 1880 Ma) in Wyoming (Spall 1971a) was assumed to differ from the curve derived from the Canadian Shield because of relative movement between the two regions (Donaldson *et al.* 1973). Other northerly poles rejected are those for the Marathon dykes, 1810 Ma, (Fahrig *et al.* 1965) and the Sudbury lopolith, 1704 Ma (Hood 1961, Sopher 1963, Larochelle 1969, Fairbairn *et al.* 1968). Recent results for the Otish gabbro, > 1790 Ma, (Fahrig & Chown 1973) and for the Spanish River Complex, 1790 Ma, (Robertson & Watkinson 1974) also favour a more northerly pole path at about 1800 Ma. Piper (1974) has shown that the a.p.w. curves from Africa and North America can be superimposed over their whole lengths prior to 1000 Ma with the two continents lying in juxtaposition. When this is done the combined data from the two regions indicates an a.p.w. curve more complicated than those proposed either by Spall or by Irving and his colleagues in that a complete loop is present between 2000 Ma and about 1750 Ma.

The mean pole from the Scourie dykes agrees with Spall's curve but not Irving & Park's. This pole also matches the return portion of Piper's loop in close agreement with the suggested age of 1800 Ma. The pole from Dyke 7, assuming magnetization at 1440 Ma, agrees well with Spall's and Irving & Park's a.p.w.; indeed this is a well documented region. Interpretation of its magnetization in terms of the older age of 1860 Ma would also fit Piper's 2000–1750 Ma loop.

The good fit of the Lewisian results to published polar wander paths tends to confirm that the Atlantic was closed in the Proterozoic and that north-west Scotland was indeed part of the North Atlantic craton (Bridgwater *et al.* 1973).

TABLE 3. *Scourie Dyke Poles*

	Before closing the Atlantic				After closing the Atlantic*	
	Lat(°N)	Long(°E)	dp	dm	Lat(°N)	Long(°E)
Mean of the group of 9 dykes	37.3	274.8	6.9	9.6	38.1	236.3
Dyke 7 (reversed)	7.9	250.8	2.2	4.4	8.3	212.6

* Using the reconstruction of Bullard et al. 1965.

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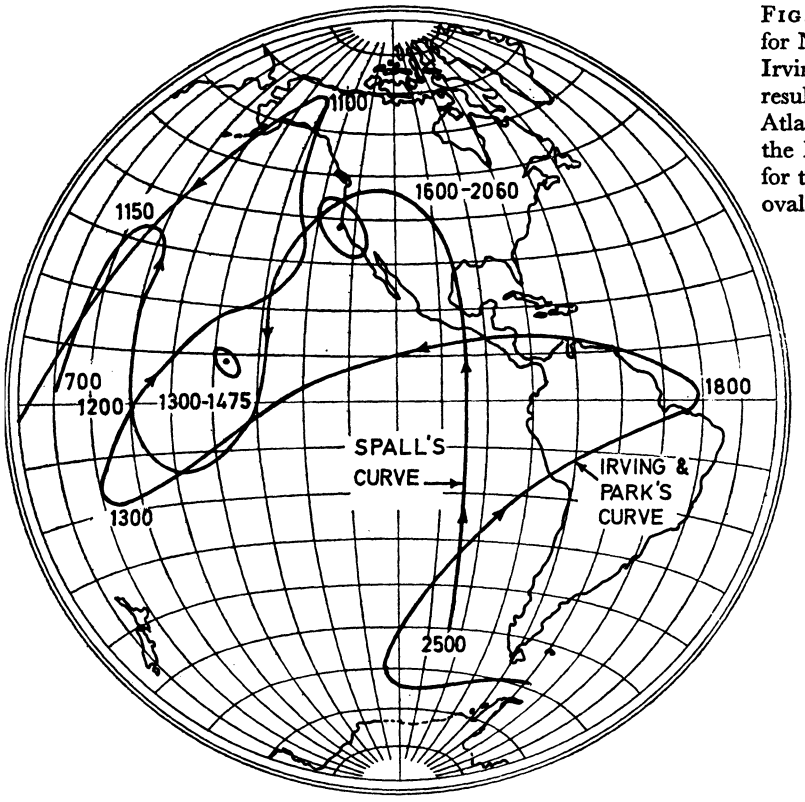


FIG. 7. Apparent polar wander curves for North American rocks (Spall 1971b, Irving & Park 1972). The Scourie dyke results are shown after closure of the Atlantic Ocean (Bullard *et al.* 1965); the large oval surrounds the mean pole for the group of 9 dykes whilst the small oval represents Dyke 7 (reversed).

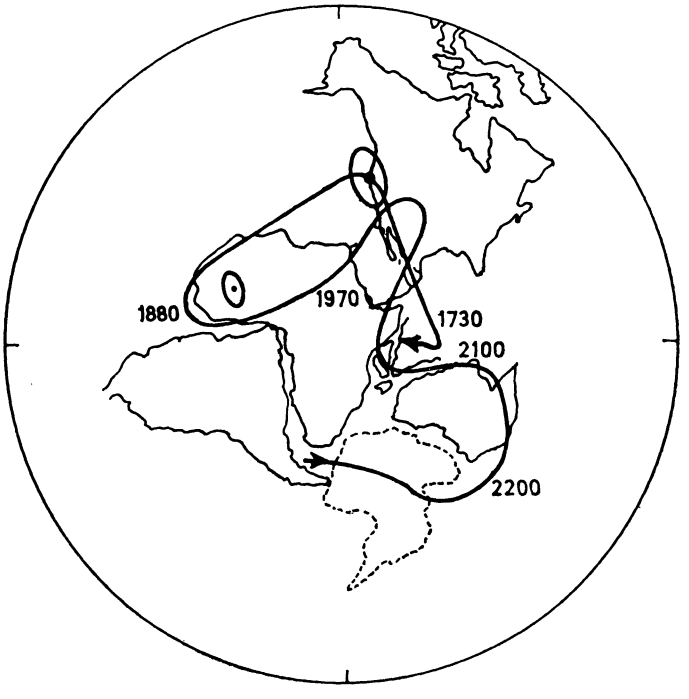


FIG. 8.

The apparent polar wander path of Piper (1974), showing the Scourie Dyke results as in Fig. 7.

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DISCUSSION

DR A. E. MUSSETT observed that little was known about secular variation in Precambrian times but, to judge by its behaviour of the last few thousand years it should be averaged out if the samples acquired their magnetism in a period of more than a few thousand years. Thus a previous comment that the data may represent a 'spot' value of the field is to re-invoke the old idea of catastrophism with a vengeance. Surely the speaker does not believe magnetization could have taken place in so short a time?

The AUTHOR replied: I believe that there were limited differences in the times of magnetization of the stable rock units. I speculate that these time differences were of the order of tens of millions of years. I have suggested the possibility that four of the dykes having high precision parameters may have each been magnetized by specific shocks; if this is correct, 'spot' values of the field would have been recorded in these cases. This may be true for the remaining stable units, or secular variation may have been averaged out by shocks occurring over a period of time.